Roydon Tay Kaiying

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EDUCATION

National University of Singapore **Aug 2023 – Present**
Bachelor of Science | GPA: 4.58/5.00

- Major in Data Science and Analytics, Minor in Computer Science
- Relevant Coursework: Machine learning, Natural Language Processing, Big Data Systems for Data Science, Mathematical Statistics, Data Structures and Algorithms, Database Management

SKILLS

- Techstack: Python, PyTorch, Scikit-learn, Pandas, Spark, Java, R, SQL
- Dev Tools: Git, GitHub, FastAPI, Docker, Cloud services (Azure, AWS)

WORK EXPERIENCE

DSO, AI Researcher Intern **May 2026 – Present**

- Researching a unified framework for LLM self-evolution loops in agentic harness engineering, combining adaptive population-based evolution with trace-based learning strategies
- Engineered experimental pipelines and executed rigorous ablation studies to quantify the impact of agentic framework components, benchmarking performance across public datasets

Shopee, Algorithm Engineer Intern **Jan 2026 – May 2026**

- Implemented and iterated upon cutting-edge recommendation model architectures from papers to Shopee's customer service chatbot, delivered measurable gains in online A/B tests
- Co-authored CIKM 2026 Applied Research Track paper submission on AuGR, a novel Generative-Ranking Joint Training framework for chatbot intent recommendation
- Performed data analysis on live traffic data to identify and resolve issues with online models
- Experimented with RQ-VAE models for generation of discrete semantic ids from dense embeddings to better integrate diverse data streams into production models

Synapxe, AI Engineer Intern **Dec 2024 – Jul 2025**

- Developed HealthHub AI chatbot to streamline health-related inquiries, leveraging RAG architecture to provide accurate, multi-lingual assistance to a diverse user base
- Spearheaded the design and deployment of an agentic chatbot capable of vaccination eligibility checking and appointment booking, showcased at AI Accelerate 2025 Conference
- Integrated automated evaluation pipelines and trace observability into LLM workflows to accelerate iterative prompt and context engineering

PSA BDP, Data Science Intern **May 2024 – Aug 2024**

- Developed text vector similarity retrieval and Reranker based search tool for Harmonized Tariff System (HTS) classification
- Identified major roadblocks affecting classification accuracy, designed LLM-powered data validation flow to improve input data consistency and prediction result recall

ADDITIONAL PROJECTS

NUS Statistics and Data Science Society, Workshops Executive **Aug 2023 – May 2025**

- Conducted workshops for fellow students discussing data science concepts and skills, such as SQL, Machine Learning workflows and exploratory data analysis methods

MediLite: Layperson Friendly Medical Knowledge Chatbot

- Utilized supervised finetuning and GRPO reinforcement learning on open-sourced LLMs with parameter efficient methods to train LLM to generate accurate and layperson friendly responses

Computer Vision for Liquid Volume Estimation

- Designed an end-to-end liquid volume prediction pipeline integrating object detection, image segmentation and deep learning vision models, achieving up to 0.91 R-squared in evaluations